

A TRIZ-based product-service design approach for developing innovative products

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ABSTRACT

As recent ICT-enabled innovations are becoming increasingly popular, there will inevitably be a need for enterprises to create innovative products and services to help improve people's daily lives. Although current product-service systems (PSS) envision a future in coupling products and services to co-create more satisfying value for both customers and enterprises, most recent studies focus on a product's service system design in the context of its use phase rather than a methodical support for implementing entire PSS design. This research aims to address the shifting interest of research and practices in Smart PSS from the viewpoint of product design and development (PDD). It presents a TRIZ-based product-service design approach to assist designers/engineers in developing innovative cyber-physical products (CPP) for use by customers. In the context of COVID-19 pandemic, an empirical study concerning a novel face mask design was conducted to verify the applicability and validity of the proposed design approach. The theoretical frameworks and practical implications of the proposed approach are also discussed.

1. Introduction

Product design and development (PDD) is a series of complex processes and activities to create products fitting customers' requirements and maintaining the competitiveness of enterprises in the market. It is highly relevant to industrial and technological innovations as well as human lifestyle needs. How to rapidly develop a product and launch it to the market has been a critical issue since the 1980s. Industry and academia have paid more attention to develop new approaches which focus on minimizing PDD time through parallelizing all possible activities in the overall PDD processes. Concurrent engineering (CE) (also known as simultaneous engineering) was emerged as an effective technique in place of traditional sequential engineering to help industries develop a product by integrating design, development, and manufacturing processes simultaneously (Smith, 1997). This technique has been proven to be capable of reducing time to market and life-cycle costs in a PDD task (Shahrokhi, Bernard, & Shidpour, 2011; Wang, Chen, Hu, & Huang, 2010). Incorporating customers' preferences into the CE technique has also been reported in the literature (Wang & Tseng, 2008). However, Valle and Vázquez-Bustelo (2009) argued that enterprises should analyze the characteristics of their innovation process and prioritize their business objectives in the context of using CE. Although the

concept of integrating CE with the supply chain design had been presented in the late 1990s (Fine, 1998; Fine, Golany, & Nasereldin, 2005), it is still insufficient for supporting the PDD performance in terms of the entire product lifecycle perspective.

In addition to the improvement of cost- and time-efficiency, human considerations have been systematically involved in PDD processes for many decades. A variety of sophisticated theories and methodologies have been proposed and employed in the literature, such as human factors and ergonomics (HFE) used for improving human performance, reducing human error, increasing productivity, and enhancing safety and comfort (Salvendy, 1997); Kansei engineering (KE) developed as a customer-oriented technology for PDD through translating customer's feelings into concrete product parameters (Nagamachi, 1995); user-centered design (UCD) regarded as a multi-disciplinary activity in which designers/engineers focus on the users and their needs in each phase of the PDD process (Vredenberg, Isensee, & Righi, 2001); and, universal design (UD) which describes the concept of PDD to be usable by all people, to the greatest extent possible, without the need for specialized design (Mace, 1985). Beyond interacting with physical objects, human-computer interaction (HCI) explores the design and use of computer technology focusing on the interaction between humans (the users) and computers (Helander, Landauer, & Prabhu, 1997);

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interaction design (IxD) helps designers/engineers to create interactive digital products, environments, systems, and services (Sharp, Rogers, & Preece, 2007); usability engineering (UE) assists designers/engineers in building bridges, providing links, and making connections between user abilities and computer hardware/software possibilities (Nielsen, 1993); and, user experience (UX) describes the consequence of a user's internal state, the features of the designed system, and the context or the environment within which the interaction occurs (Hassenzahl & Tractinsky, 2006).

On the other hand, environmental concerns have become strategically important due to the dramatic growth of environmental problems and public awareness of environmental issues. There have been a significant number of strategies and approaches in response to the concerns including, for example, green design that treats environmental attributes as PDD objectives by introducing the principle of the 3Rs: reducing, reusing and recycling (Dowie, 1994); eco-design which integrates environmental considerations into the PDD process (Brezet & van Hemel, 1997); design for environment (DfE) which takes a life-cycle approach to PDD, considering such novel concerns as environmental consequences, human health and safety (Fiksel, 1996); sustainable design which intends to reduce negative environmental impact completely through proficient and sensitive design (McLennan, 2004); cradle-to-cradle design (C2C) that presents an alternative PDD concept to the zero emission and eco-efficiency strategies, seeking to reduce the unintended negative consequences of production and consumption processes, and incorporating social, economic, and environmental benefits in products and services (Braungart, McDonough, & Bollinger, 2007); design for sustainability (D4S) that builds on the work of eco-design to comprise economic and social concerns and outlines methodologies for making sustainable improvements to products by applying lifecycle thinking components (Clark, Kosoris, Hong, & Crul, 2009); and more recently, circular design which focuses on minimizing the use of primary raw materials to curtail a value loss embedded in the products and materials, by keeping them in a closed-loop circulation mode (Medkova & Fifield, 2016).

Moreover, various life cycle approaches have been employed to assist designers/engineers in assessing the sustainability of a PDD task (Gundes, 2016), such as life cycle assessment/analysis (LCA), life cycle costing (LCC), life cycle sustainability analysis (LCSA), and life cycle engineering (LCE). Some visualization tools and checklists are also available for use in PDD processes to help achieve eco-innovation through identifying potential environmental impacts, such as the life-cycle design strategy (LiDS) wheel (Brezet, 1996), EcoCompass (Fussler & James, 1996), product ideas tree (PIT) diagram (Jones, Stanton, & Harrison, 2001), etc. (for more detailed information, please refer to Byggeth & Hochschorner, 2006). Otherwise, the family of Design for X (DFX, also known as Design for Excellent) provides cross-functional insights into PDD implementation for both engineering (e.g., Design for manufacturing/assembly/reliability/quality) and environmental (e.g., Design for disassembly/reuse/recycling/maintainability) performances (Kuo, Huang, & Zhang, 2001). Sustainability has been a critical issue in the context of PDD toward environmental concerns since the late 1980s. Ceschin and Gaziulusoy (2016) provided a systematic overview of the design for sustainability domain and categorized the design approaches under four innovation levels: product, product-service system, spatio-social system, and socio-technical system. They also indicated that the design for sustainability has increasingly evolved from a technical and product-centric focus to an extensive system level change in which sustainability is perceived as a socio-technical challenge.

Developing service offerings to facilitate customer value co-creation processes has become an important issue in a PDD task beyond fitting customers' requirements or incorporating customers' preferences (Wetter-Edman et al., 2014). Many researchers argued that industrial and technological innovations can yield an improvement in products' environmental performance, but such an improvement can often be negatively counterbalanced by increasing a certain degree of

consumption (Binswanger, 2001; Brookes, 2000). In the last decade, there has been an increasing trend for enterprises to shift their business strategies from designing, developing, and manufacturing products to selling, delivering, and innovating services (Gebauer, Edvardsson, Gustafsson, & Witell, 2010; Kowalkowski, Gebauer, & Oliva, 2017; Qi, Mao, Zhang, & Guo, 2020; Ulaga & Reinartz, 2011). Product-service systems (PSS) can be considered a novel concept for enterprises to improve their sustainability performance with the potential to minimize environmental impacts of both production and consumption. Manzini and Vezzoli (2002) defined PSS as an innovation strategy of shifting the business focus from merely designing and selling physical products to innovating and delivering a system of products and services. Current PSSs are endowed with a market proposition that expands a product's functionality by unifying additional services to create value for customers and further build positive customer relationships for enterprises (Baines, Lightfoot, Benedettini, & Kay, 2009; Beuren, Ferreira, & Miguel, 2013; Kuijken, Gemser, & Wijnberg, 2017). They are complex artefacts composed of products, services, business models, and a network of stakeholders who produce, deliver, and manage the PSS (Mont, 2002). The design of PSS, therefore, requires a systematic approach to couple with the relevant components simultaneously.

Due to the common PSS implementation is likely to focus on the use phase of a product's life cycle (Mont, 2002), current PSS design methodologies mostly support the service design processes, and do not adequately provide engineering solutions to facilitate the design of a physical product (Maussang, Zwolinski, & Brissaud, 2009). Trevisan, Lelah, and Brissaud (2015) further explained the existing challenge of coupling products and services in design processes as these two tasks have respective insights and use specific techniques which seems to be difficult to integrate, let alone the business models and/or the network of stakeholders. Many researchers have conducted literature reviews on PSS; most of them found that the research in this area is still dominated by theoretical work, and methodological development, exclusive design of PSS, and empirical research/practice cases are required (Annarelli, Battistella, & Nonino, 2016; Beuren et al., 2013; Salwin, Kraslawski, & Lipiak, 2020). Qu, Yu, Chen, Chu, and Tian (2016) pointed out that PSS design methodologies can be classified into 6 types, namely customer perspective, modeling techniques, visualization methods, modularity, TRIZ, and system dynamics. Over the past few decades, TRIZ has been widely and successfully employed by researchers and practitioners in fields across different disciplines (e.g., Boavida, Navas, Godina, Carvalho, & Hasegawa, 2020; Chen & Liu, 2001; Rupani, Lotfi, Nejat, Abu Bakar, & Othman, 2017; Wu, Zhou, & Kong, 2020). It has the potential to address complex PDD issues involving product, service, and sustainability aspects. Accordingly, this research presents a TRIZ-based product-service design approach to assist designers/engineers in developing innovative products for use by customers. The rest of this paper is structured as follows. Section 2 introduces the material and methods. Section 3 elaborates the proposed approach, while Section 4 presents the results of applying the approach to developing a novel product for use against the COVID-19 pandemic. A discussion is provided in Section 5, and the following conclusions and recommendations for future research are given in Section 6.

2. Material and methods

TRIZ (Theory of Inventive Problem Solving) was initially developed by Althuller and his colleagues in the mid-1940s. It is a systematic and practical methodology for designers/engineers to analyze inventive problems and further resolve them in a strategy-directed manner (Alshuller, 1996). The philosophy of TRIZ has evolved from original 4 pillars (ideality, resources, functionality, and contradiction) and an additional fifth one (space/time/interface) to updated 7 pillars (plus system transfer and system transition) (Sheu, Chiu, & Cayard, 2020), which shows that this methodology is not old-fashioned but continuously progressive. The fundamental of TRIZ is to eliminate

contradictions with available resources to improve functionality as well as increase the ideality of a technological system. TRIZ necessitates the interconversion between factual domains and TRIZ domains, and suggests a four-stage process for solving problems: (1) finding factual problems; (2) formulating as TRIZ generic problems; (3) generating TRIZ generic solutions; and, (4) converting to specific solutions. In TRIZ domains, generic problems can be formulated by using TRIZ analytical tools, such as project definition, functional modeling/analysis, patterns of technological evolution, ideality and ideal final result (IFR) analysis, S-curve analysis, multi-screen diagram/9 windows, and algorithm for inventive problem solving (ARIZ). To generate generic solutions, designers/engineers have to categorize the TRIZ problems into classifications and select proper tools for solving the problems. For each problem classification, several knowledge-based tools can be used to resolve the problems, such as 39 engineering parameters and the contradiction matrix, 40 inventive principles for technical contradictions, 11 separation principles for physical contradictions, 76 standard solutions for substance-field (Su-field) analysis, 8 evolutionary patterns for forecasting technology trends, and the effects database for facilitating the use of TRIZ principles (Ilevbare, Probert, & Phaal, 2013; Kaplan, 1996). Taking advantage of the corresponding tools, many possible TRIZ solutions can be generated and then are converted into specific solutions to address the factual problems for further engineering development. With its excellent effectiveness and applicability, TRIZ has been extensively employed by both academia and industry over the past few decades (Aguilar-Lasserre et al., 2020; Chechurin & Borgianni, 2016; Delgado-Maciel, Cortés-Robles, Sánchez-Ramírez, García-Alcaraz, & Méndez-Contreras, 2020; Ekmekci & Koksall, 2019), not only in traditional engineering fields but also in business and management (Souckov, 2019), services (Chai, Zhang, & Tan, 2005; Lee, Zhao, & Lee, 2019; Lee, Chen, Li, & Shie, 2020), maintenance (Vaneker & van Diepen, 2016), green supply chain (Moussa, Rasovska, Dubois, De Guio, & Benmoussa, 2017), and sustainable innovation (D'Anna & Cascini, 2011). Although TRIZ has great potential to innovate PSS design, there has been relatively little research on how to apply TRIZ to designing PSS (de J. Pacheco, ten Caten, Navas, Jung, & Cruz-Machado, 2016).

Literature provides multiple definitions on PSS with different emphasized viewpoints, such as business strategies, marketing, services, and product management (Baines, Lightfoot, & Evans, 2007; Morelli, 2006). PSS ontology should identify such characteristics as requirements, product and service contents, stakeholders, design processes, lifecycle phases, business models and support systems (Vasanth, Roy, Lelah, & Brissaud, 2012). From the lifecycle perspective, PSS design could be considered an expansion of CE that couples with products, services, business models, and sustainability to coordinate the overall PDD processes and activities to fulfill customers' requirements while minimizing environmental impact. As aforementioned, PDD is highly relevant to industrial and technological innovations as well as human lifestyle needs. After entering the 21st century, the world has changed dramatically and new industrial technologies and business models are emerging increasingly and rapidly. E-commerce, social networks, mobile platforms and apps, robotics, artificial intelligence (AI), extended reality (XR), and cloud computing are changing our lifestyles as well as influencing the social, economic, and environmental conditions. In this context, a new PDD paradigm is needed, and PSSs have also evolved from Internet-based PSS (conventional PSS), IoT-enabled PSS, to current Smart PSS (Cong, Chen, Zheng, Li, & Wang, 2020; Zheng, Wang, Chen, & Khoo, 2019). Smart PSSs are enabled by the state-of-the-art information and communication technologies (ICTs) (Valencia, Mugge, Schoormans, & Schifferstein, 2015) and have attracted much attention among academics in recent years (Chowdhury, Haftor, & Pashkevich, 2018; Li, Wang, Chen, & Zheng, 2021; Liu, Ming, Qiu, Qu, & Zhang, 2020; Pan, Zhong, & Qu, 2019; Watanabe, Okuma, & Takenaka, 2020). Cong et al. (2020) conducted a systematic literature review on existing engineering design methodologies, including TRIZ, quality function deployment (QFD), Kansei engineering (KE), user-centered

design (UCD), axiomatic design (AD), blueprint design, adaptable design, mass customization (MC), and functional-behavior-structure (FBS), and selected 101 representative items in relation to the Smart PSS development. They found that there is still lack of a fundamental design methodology for the development of Smart PSS. The difficulty of developing Smart PSS design methodologies could be due to its broad definitions in what we exactly consider to be a smart PSS, a product-based service system, a service-oriented product system, or others. A number of questions regarding the Smart PSS design remain to be addressed.

To fill the gap, the aims and scope of the Smart PSS design should be clarified. According to the author's previous work, Smart PSS could be regarded as cyber-physical products (CPP) which are described as a physical product that offers ICT-enabled services via either a built-in or an external network connecting device, allowing users to use the product functionality and communicate information during its use phase for acquiring complementary service support (Chou, 2020). More succinctly, CPPs are considered a new type of consumer products (physical entities) embedded with smart components and connectivity (cyber-space) to augment the capabilities and value of products through the acquisition of complementary services. This description is similar to some other authors who argue Smart PSS fundamentally composed of smart, connected product and its generated digital services (e.g., Kuhlénkötter et al., 2017; Valencia et al., 2015; Zheng, Lin, Chen, & Xu, 2018), which means that the provided services are central to the product. The term CPP is coined by combining the concepts of cyber-physical systems (CPS) and Internet of Things (IoT) into a PDD. In addition to physical products, the serviceability of products is a key feature of the CPP, which could help develop a network-based business model to create a value co-creation platform for both customers and enterprises. In such a service-enabled product, customers can acquire complementary services during the product use phase, while enterprises can create more profit from both the product selling and service offering. With the clear definition and definite contents of CPP, the approach focuses on the design and development of a physical product incorporating its potential service support to satisfy customers' requirements while minimizing environmental impact. There are 4 pillars contained in the framework of the product-service design approach, namely functionality, serviceability, usability, and sustainability. The implementation steps of the proposed approach are detailed in the following section.

3. Proposed approach

This study focuses on PDD to develop innovative cyber-physical products (CPP) for use by customers. Based on the material and methods given in Section 2, the framework of the TRIZ-based design approach is built under a horizontal timeline of life cycle stages and vertical flowcharts of the design processes as shown in Fig. 1. The design processes are illustrated step by step as follows:

3.1. Step 1. Identification of needs

PDD aims to develop products fitting customers' requirements and enhancing enterprises' competitive advantage in the marketplace. Both the customer needs and target markets must be identified in order to understand the aim of the PDD. In addition to functionality and other product features, designers/engineers have to ponder how the target product can be serviceable (capable of or being of service) in a commercial networked manner to maintain, extend, and/or enhance its functionality. This is the essence of designing cyber-physical products (CPP).

3.2. Step 2. Problem definition

PDD is a series of creative activities relied heavily on exploring innovative solutions within our daily lives through an organized

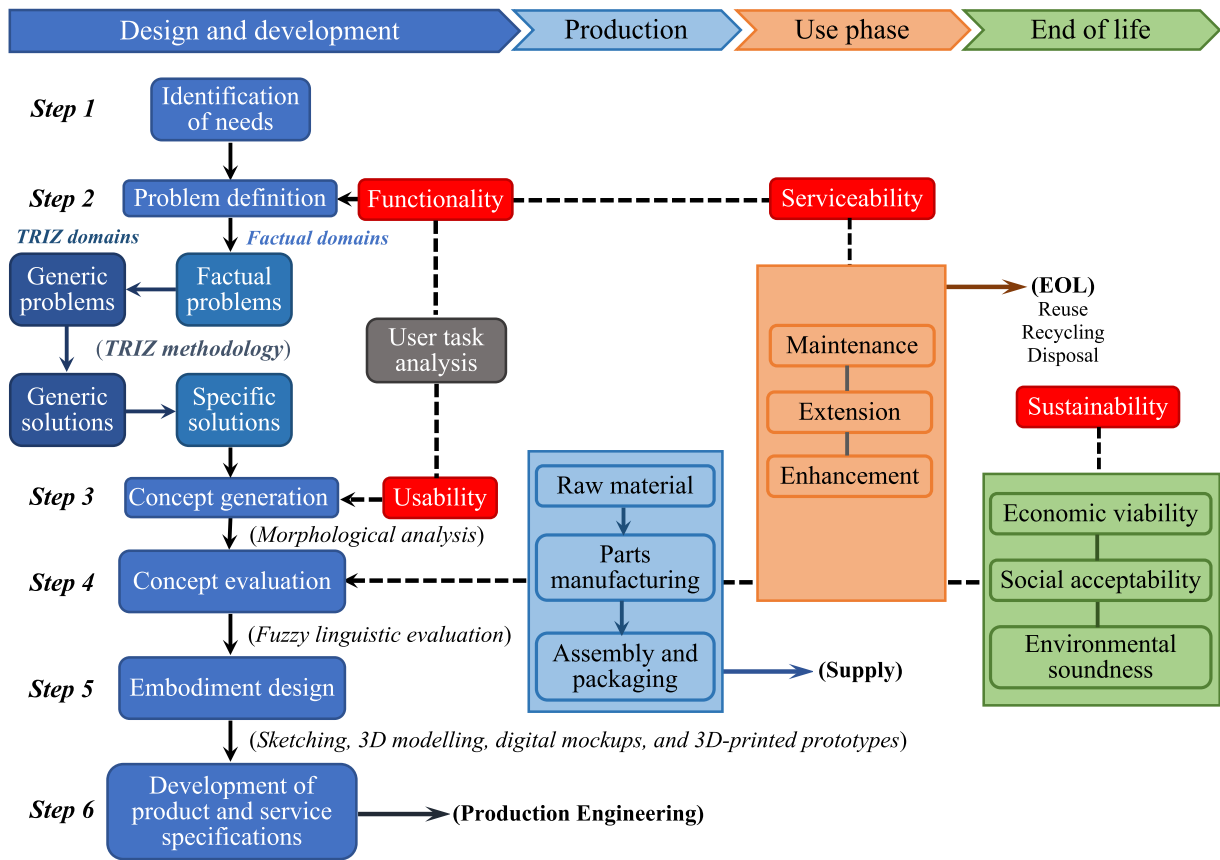


Fig. 1. Framework of the TRIZ-based product-service design approach.

combination of finding, analyzing, and resolving problems. Contradiction problems often occur in not only product design but also service design processes (Song & Sakao, 2016). Problem definition is aimed at finding factual problems involved in the PDD task. These factual problems must be formulated as TRIZ generic problems by using TRIZ analytical tools mentioned above.

3.3. Step 3. Concept generation

Early design processes were often compared to a “black box”, a phenomenon with known inputs and outputs but with an unknown internal mechanism, which implies that designers/engineers know what it does and not by how it does it. To fill the gap, TRIZ methodology provides designers/engineers with a predictable and strategy-directed problem-solving process to generate TRIZ generic solutions. In addition to conventional TRIZ knowledge-based tools for use in engineering and technical fields, there have been a number of modified and extended tools that can be applied to non-technical areas, such as the modified inventive principles for service operations (Zhang, Chai, & Tan, 2003), the service TRIZ contradiction matrix for service quality design (Shahin & Pourhamidi, 2011), the contradiction parameter options for service system design (Jiang, Sun, & Cheng, 2013), and the interpretation of the 40 inventive principles for performing services (Gazem & Rahman, 2014). Moreover, the interaction design guidelines (IDGL) (Filippi & Barattin, 2015) and the iTREE/ECO-TRIZ guidelines (Russo, Rizzi, & Montelisciani, 2014) are available for designers/engineers to connect the TRIZ generic solutions with usability and sustainability parameters.

To ensure product and service quality under the physical and digital environments, user task analysis should also be conducted in this step. The key point of user task analysis is to understand how users can perform and complete relevant tasks to achieve their goals, providing them with access to intuitively interact with the CPP with positive UX

and without excessive mental workload. Through TRIZ operations, many possible generic solutions can be generated. As these solutions are suggested by TRIZ generic guidelines, principles, and/or parameters built in the TRIZ toolkit, they are abstract concepts and should be converted into specific concrete solutions. Based on the factual problems and their corresponding specific solutions, the product and service can be decomposed respectively into different constituting components. These components are used as option parameters to perform the morphological analysis. A large number of design concepts can be generated through permuting and composing these option parameters. After completing a preliminary assessment, a set of alternative design concepts can be derived.

3.4. Step 4. Concept evaluation

Mont (2002) argued the necessity of practical implementation and evaluation of economic, environmental, and social consequences for PSS design. Concept evaluation aims to identify promising design concepts on the basis of the preferences of the evaluators. The economic, environmental, and social components of sustainability are used as criteria to assess the alternative design concepts, where each criterion comprises three sub-criteria as listed in Table 1.

The fuzzy linguistic approach with the aggregation operator of fuzzy weighted average (FWA) is used to evaluate the alternative design concepts. Given a set of linguistic rating variables $\tilde{R} = \tilde{r}_{ij}$, and a set of linguistic weight variables $\tilde{W} = \tilde{w}_{ij}$, the formula of FWA is expressed as follows:

Table 1
List of sustainability criteria for the concept evaluation.

Criterion	Sub-criterion	Description
C ₁ : Economic viability	Manufacturing costs (c ₁₁)	The design can minimize manufacturing costs.
	Assembly/packaging costs (c ₁₂)	The design can minimize assembly and packaging costs.
	Maintenance costs (c ₁₃)	The design can reduce maintenance costs.
C ₂ : Social acceptability	Quality of life (c ₂₁)	The design can improve the quality of life.
	Health and well-being (c ₂₂)	The design can promote health and well-being to humans.
	Safety and security (c ₂₃)	The design can ensure safety and security throughout the product lifecycle.
C ₃ : Environmental soundness	Materials used (c ₃₁)	The design can minimize the use of materials.
	Energy saving (c ₃₂)	The design can conserve energy during production and use phase.
	Waste reduction (c ₃₃)	The design can expand product lifespan, reduce waste, and further harmonize the end-of-life (EOL) treatments.

$$D_k(\tilde{R}, \tilde{W}) = \frac{\sum_{i=1}^m \sum_{j=1}^n \tilde{r}_{ij} \cdot \tilde{w}_{ij}}{\sum_{i=1}^m \sum_{j=1}^n \tilde{w}_{ij}} = \frac{\int_{\alpha=0.063}^1 (\tilde{r}_{11} \times \tilde{w}_{11})_{\alpha} + \dots + (\tilde{r}_{ij} \times \tilde{w}_{ij})_{\alpha} + \dots + (\tilde{r}_{mn} \times \tilde{w}_{mn})_{\alpha}}{\int_{\alpha=0.063}^1 (\tilde{w}_{11})_{\alpha} + \dots + (\tilde{w}_{ij})_{\alpha} + \dots + (\tilde{w}_{mn})_{\alpha}} \quad (1)$$

where i is the number of criteria; j is the number of sub-criteria; k is the number of alternative design concepts; and, $D_k(\tilde{R}, \tilde{W})$ represents the overall desirability of alternative design concept k .

Linguistic variables $\tilde{r}_{ij}$ and $\tilde{w}_{ij}$ are Gaussian fuzzy numbers, where the fuzzy membership function is $\mu_A(x) = e^{-\frac{1}{2}(x-c/\sigma)^2}$; c is the center (i.e., mean) and σ is the width (i.e., standard deviation) of the fuzzy set. As shown in Fig. 2, a 7-point rating scale (VL, L, ML, M, MH, H, and VH) described by the continuous Gaussian membership functions (GMFs) (where $2\sigma^2 = 0.01$) is used to evaluate the alternative design concepts. The aggregation with continuous α -cuts ($0.063 \leq \alpha \leq 1$) is based on the

arithmetic operations (addition, multiplication, and division) on closed intervals, as defined below:

Let $r_{\alpha} = [a, b]$ and $w_{\alpha} = [c, d]$ be the two closed intervals of real numbers, then

- $(\tilde{r} \times \tilde{w})_{\alpha} = \tilde{r}_{\alpha} \cdot \tilde{w}_{\alpha} = [a, b] \cdot [c, d] = [\wedge(ac, ad, bc, bd), \vee(ac, ad, bc, bd)] = [e, f] = \tilde{z}_{\alpha}$;
- $(\tilde{z}_1)_{\alpha} + (\tilde{z}_2)_{\alpha} + \dots + (\tilde{z}_m)_{\alpha} = [e_1, f_1] + [e_2, f_2] + \dots + [e_m, f_m] = [(e_1 + e_2 + \dots + e_m), (f_1 + f_2 + \dots + f_m)] = [\varepsilon, \eta]$;
- $(\tilde{w}_1)_{\alpha} + (\tilde{w}_2)_{\alpha} + \dots + (\tilde{w}_m)_{\alpha} = [c_1, d_1] + [c_2, d_2] + \dots + [c_m, d_m] = [(c_1 + c_2 + \dots + c_m), (d_1 + d_2 + \dots + d_m)] = [\rho, \sigma]$; and provided that $0 \notin [\rho, \sigma]$
- $\frac{[\varepsilon, \eta]}{[\rho, \sigma]} = [\varepsilon, \eta] \cdot \left[\frac{1}{\sigma}, \frac{1}{\rho}\right] = \left[\wedge\left(\frac{\varepsilon}{\rho}, \frac{\varepsilon}{\sigma}, \frac{\eta}{\rho}, \frac{\eta}{\sigma}\right), \vee\left(\frac{\varepsilon}{\rho}, \frac{\varepsilon}{\sigma}, \frac{\eta}{\rho}, \frac{\eta}{\sigma}\right)\right]$

The result of the arithmetic operations is a crisp set (interval) that represents the α -cut of the fuzzy set obtained through operating on fuzzy numbers $\tilde{r}_{ij}$ and $\tilde{w}_{ij}$. Taking advantage of the FWA aggregation operations, the family of α -cuts can be defined as the resulting membership function of the evaluated alternative design concept, representing as a convex and normalized fuzzy set, $D_k(x)$, which is also classified as a fuzzy number. In order to derive a quantitative value, $\bar{D}_k$, the defuzzification method of center-of-gravity (COG) is used as given in Eq. (2); the higher the $\bar{D}_k$ value, the better is the evaluated alternative design concept. An example of the aggregation and defuzzification result is also shown in Fig. 2.

$$\bar{D}_k = \frac{\int_s^t D_k(x) \cdot x dx}{\int_s^t D_k(x) dx} \quad (2)$$

where $D_k(x)$ represents the resulting membership function of the alternative design concept k ; s and t are the respective lower and upper limits of the support of the fuzzy number.

3.5. Step 5. Embodiment design

According to the concept evaluation result, the most desirable design concept must be embodied to realize the product concept associated with its service scheme. Since CPPs consist of both tangible products and intangible services, the embodiment design focuses on outlining and

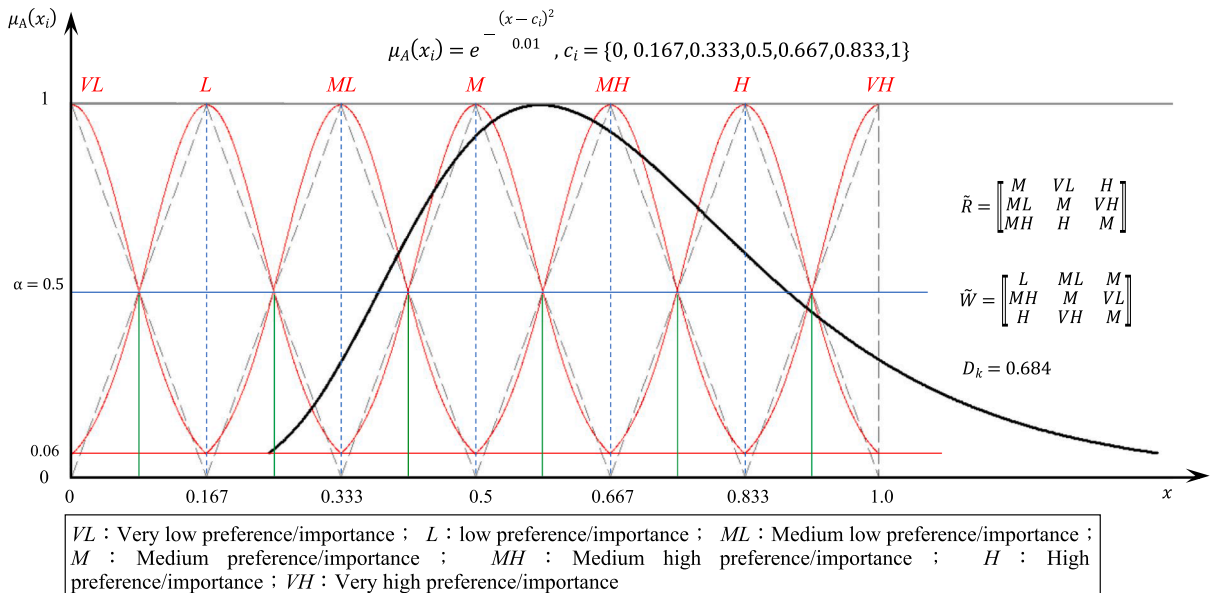


Fig. 2. Seven-point scales characterized by continuous Gaussian membership functions.

allocating physical components to the functions of a product, and scheming out interactive contents to its complementary services. Visualization techniques for engineering development are used to embody product concepts and verify the feasibility of the CPP design, including sketching, 3D modelling, digital mockups, and 3D-printed prototypes.

3.6. Step 6. Development of product and service specifications

The product specifications are developed according to the final PDD results. They describe and state the intended use, operating conditions, specific features and characteristics, and performance/capability of the product. The service specifications are also determined to describe the methods and schemes for the service support, while the features of user interfaces are elaborated to specify how to jointly and consistently use the product and its service support.

4. Results

Coronavirus disease 2019 (also known as the COVID-19) is an infectious disease first detected in December 2019, and then has resulted in an ongoing global pandemic (Hui et al., 2020). As of mid-November 2020, more than 53.4 million cases have been reported across about 218 countries, areas or territories, resulting in more than 1.3 million deaths. Although there are so many COVID-19 vaccines in development, so far there are no highly effective vaccines nor specific antiviral treatments for COVID-19. The COVID-19 outbreak affects not only the public health but also global politics, economics, society, and environment (Chakraborty & Maity, 2020). Accordingly, this section presents the results of applying the proposed approach to developing a novel cyber-physical product for use against the COVID-19 pandemic.

4.1. Identification of needs

In addition to vaccines and medical treatments, individual prophylaxis is a vital measure against the COVID-19 pandemic, as the virus is primarily spread between people during close contact, most often through small droplets emitted by coughing, sneezing, and talking. Cirrincione, Plescia, Ledda, Rapisarda, and Martorana (2020) argued that personal protective equipment (PPE) must be used to reduce the risk of spreading viruses in the environment, whenever there is potentially close contact with a suspect case or even with a potentially infected person who does not wear a surgical mask. The World Health Organization (WHO, 2020) put forward advice on the use of masks in the context of COVID-19. Many European nations and several US states have also issued advisories to wear a mask in certain public settings, covering in enclosed spaces where social distancing is not always possible. Mask-wearing has been perceived as a prosocial behavior that people act to benefit others rather than themselves within their community. However, some people around the world do not like to wear a face mask in public. The reasons of refusal are quite complicated; for example, cultural norms and social pressure may restrict mask-wearing in public, which interprets why masking has been restrained in the West. Except for cultural stereotypes, social conventions, personal faith and belief, and political guidance confusion, the barrier to mask-wearing can be resolved through novel product and service design to assist people overcome their misgivings over wearing face masks. Hence, this design case aims to develop a novel face mask product to satisfy customers' demands.

4.2. Problem definition, classification, and tool selection

Although wearing face masks in public does offer protection against the spread of COVID-19, some people still do not want to wear masks. Through reviews on public-health blogs, websites, and influencers' comments, the reasons why people refuse to wear masks are defined as the factual problems for the TRIZ implementation. These reasons are

classified according to the product's functionality and serviceability as listed below:

1. People are particularly likely to opt out of wearing masks, believing them to be "shameful" and "a sign of sickness".
2. People from certain groups may not feel comfortable while wearing a mask, particularly wearing a non-medical or home-made mask, due to concerns of racial profiling.
3. Some masks do not exactly fit someone's face.
4. Most masks make wearers feel embarrassing and unfashionable.
5. Current mask design is not pleasing and artistic.
6. Masks do keep people from seeing facial expressions and hearing voices clearly, both essential for social connection.
7. Mask wearing makes noses itch and can cause skin irritation.
8. Although masks do not deprive people of oxygen, they do make wearers harder to breathe freely.
9. Wearing a mask can get sweaty, feel stuffy and damp from the moisture in the wearer's breath and may fog up his/her glasses.
10. There have been concerns that mask wearing could engender a false sense of security with respect to other infection control measures such as social distancing and handwashing.
11. There have been shortages of face masks, and not all have been certified.
12. Standard masks were reported on the market, which can significantly reduce the protective performance.
13. It could be difficult to purchase masks due to the shortage of mask supply in the community.
14. How do I know when it is time to get/replace a new mask?
15. Masks shall be tested to ensure that the essential requirements are met.
16. Authorities also give close attention to exact techniques for wearing, removal, and disposal of face masks.

Problems 1, 2, and 4 are regarded as the administrative contradictions which occur when designers/engineers recognize the need for a change but do not know how to do it. The ARIZ can be used to resolve such abstract problems by applying a multi-step process to transforming and formulating initial problems as well as analyzing and resolving the problems. These 3 problems were transferred to an initial situation for the ARIZ operation, formulated as "How can we improve the face mask design to reduce the feelings of shame and embarrassment for wearers, as well as to make the design more stylish and fashionable?" (TRIZ generic problem 1, TP 1). Problem 3 is regarded as the imperfect functional structure (TP 2) that can be resolved using the "Functional Improvement Methods" and "TRIZ Standard Solutions" tools. The functional diagram is shown in Fig. 3.

Problems 5 and 16 are regarded as the system improvement (TP 3, TP 4) that can be resolved by using the "S-Curve Analysis/Patterns of Technological Evolution" tool. Problems 6, 8, 9, and 10 are considered the physical contradictions, in which problems 6 and 10 can be combined into one TRIZ generic problem due to the concerns of social connection and communication, while problems 8 and 9 can be combined into one TRIZ generic problem concerning the breathing resistance effect. These problems can be resolved by using the "Resolution/Separation Principles" tool and are formulated as follows:

TRIZ generic problem 5: A mask should have the characteristic of creating a physical barrier for retaining large droplets released from the mouth and nose by the wearer in public, and it should have the characteristic of allowing interaction between the wearer and people

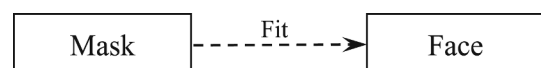


Fig. 3. Functional diagram of TRIZ generic problem 2 (TP 2).

around him/her in order to satisfy the limitation and requirement of social connection and communication. (TP 5)

TRIZ generic problem 6: A mask should have the characteristic of creating a physical barrier between the mouth and nose of the wearer in order to help reduce exposure of the wearer's saliva and respiratory secretions to others, and it should have the characteristic of allowing good ventilation in order to satisfy the limitation and requirement of reducing moisture and heat inside the mask and making wearers easier to breathe freely. (TP 6)

Problem 7 is considered a technical contradiction, while Problems 11, 12, and 15 are regarded as the technical contradictions in product serviceability. Since the latter 3 problems are in the context of production and supply of services, they were combined into one TRIZ generic problem. These problems can be resolved by using the "Contradiction Matrix/Inventive Principles" tool, and are formulated as follows:

TRIZ generic problem 7: Increasing the use of mask-wearing can protect people against the spread of COVID-19 (improving: #27 Reliability), but it can cause itchy noses and skin irritation for the wearer (deteriorating: # 31 Object-generated harmful factors). (TP 7)

TRIZ generic problem 8: Increasing mask production quickly can ensure a sufficient supply of masks (improving: # 39 Productivity), but it can reduce the quality (deteriorating: # 29 Manufacturing precision) and performance (deteriorating: # 27 Reliability) of the masks. (TP 8)

Problems 13 and 14 are regarded as the system improvement in product serviceability, which aim to improve the mask usage and supply system to enhance customer services (TP 9, TP 10). They can be resolved by using the "Multi-Screen Diagram/9 Windows" tool to help analyze the product and service system and predict the further evolution of the system.

4.3. Concept generation and preliminary assessment

The ARIZ operation for resolving the contradiction problem of TP 1 was conducted as listed in [Appendix A](#). Wearable devices are inferred as the existing resource for the system change and resolution development of mask-wearing. According to "TRIZ Separation Principles" (Separation upon condition), the face mask design should be separated from its conventional image (a medical device) and usage (wearing a PPE). By using "TRIZ Inventive Principles" (Principle 3. Local quality), the face mask design is changed from a traditional medical device to a modern wearable device (Specific solution 1).

Mask-wearing should fit above the nose and below the chin snugly, but not so tightly that it digs into the wearer's face. A face mask (also known as a surgical mask/procedure mask) mostly consists of a three-ply pleated material (made up of a melt-blown polymer placed between two layers of non-woven fabrics), which acts as the filtering composition to stop microbes from entering or exiting the mask. The masks allow wearers to expand the pleated material to cover the area from the nose to the chin, and are secured to the head with ear loops, head ties, or elastic straps. A metal band is also used as the nose bridge that allows the wearer to adjust it so the mask contours with his/her face. According to the "Functional Improvement Methods" and "TRIZ Standard Solutions", we can add a subject to create an adequate useful action for TP 2, by where a flexible frame made from an elastic memorized material is used to uphold the filtering composition, allowing the wearer to adjust the mask contoured to fit his/her face (Specific solution 2). Moreover, the flexible mask frame provides different sizes for different requirements (Specific solution 2.1). The functional improvement diagram is shown in [Fig. 4](#).

Covering the nose and mouth had been part of traditional sanitary practices against contagious diseases in early modern Europe. The face masks have evolved over the centuries, and many historical surveys can

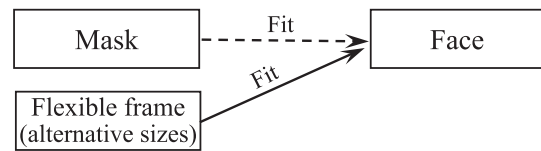


Fig. 4. Functional improvement diagram of TRIZ generic problem 2 (TP 2).

be found in the literature (e.g., [Spooner, 1967](#); [Strasser & Schlich, 2020](#)). [Fig. 5](#) presents the S-curve of the technological evolution for medical face masks. In accordance with the "Patterns of Technological Evolution" tool, TP 3 can be resolved by "increasing dynamism and controllability". Accompanied by the evolution of wearable technology in healthcare ([Aliverti, 2017](#); [Gao, Li, & Luo, 2015](#)), the face mask can be designed by combining electronic components and software to develop a novel wearable device with fashionable appearance (Specific solution 3). On the other hand, there has been a step-by-step guide on how to properly put on, wear, take off, and dispose of a face mask, announced by the World Health Organization (WHO), the Centers for Disease Control and Prevention (CDC, USA), and many infectious disease specialists. However, there are still a lot of people around the world using face masks incorrectly, let alone the use of wearable devices. TP 4 can be resolved by "increasing complexity followed by simplicity through integration". A wearable device can be considered a cyber-physical product that is worn close to and/or on the surface of the skin, where it detects, analyzes, and transmits information regarding physiological body signals, such as vital signs and ambient data. These also allow immediate biofeedback to the wearer and/or to remote operators in healthcare. The wearable device simplifies the complexity of wearable technology applications to capture or track biometric information related to breath health or fitness only, and is integrated into the novel face mask design (Specific solution 4).

A face mask should positively interact with the wearer to create a physical barrier for retaining large droplets released from the mouth and nose, but it could create a social barrier for the wearer to interact with people in public. The functional model of TP 5 is shown in [Fig. 6](#).

The contradiction of TP 5 is that a face mask should create a physical barrier and could not cause a social barrier while a wearer wears it (the time property of conflict) in public (the location property of conflict). According to "TRIZ Separation Principles" (Separation between part and a whole), the face mask design can add an LED indicator (Specific solution 5) or a microphone (Specific solution 6) to help wearers communicate with people. By using "TRIZ Inventive Principles" (Principle 15. Dynamics), the indicator can display different color signals according to the air quality index (AQI) by the wearer's immediate location to remind people around him/her (Specific solution 5.1). The AQI data are derived from the wearer's smartphone through the GPS and Bluetooth connection (Principle 2. Taking out) (Specific solution 5.2). Here the smartphone is used as an external network connecting device to maximize utilization of existing resources and increase the ideality of the system. The microphone can convert the wearer's voice into electrical signals for translating into other languages via translation apps (e.g., Google Translate, Microsoft Translator, VoiceTra, and iTranslate Voice) installed on his/her smartphone (Principle 24. Intermediary) (Specific solution 6.1). These can make more friendly connections between people and mask wearers and eliminate the pandemic and cultural misgivings about wearing a mask.

As humans breathe through the nasal passage, covering the mouth and nose does compromise the wearer's air exchange. The contradiction of TP 6 is concerned with the physical barrier and ventilation. According to "TRIZ Separation Principles" (Separation between part and a whole), the face mask design can add a micro exhaust fan as the field of mechanical energy to expel exhalation according to the wearer's respiratory cycle (Specific solution 7). The Su-field analysis models are shown in [Fig. 7](#). This can also help expel heat for reducing temperature, reduce

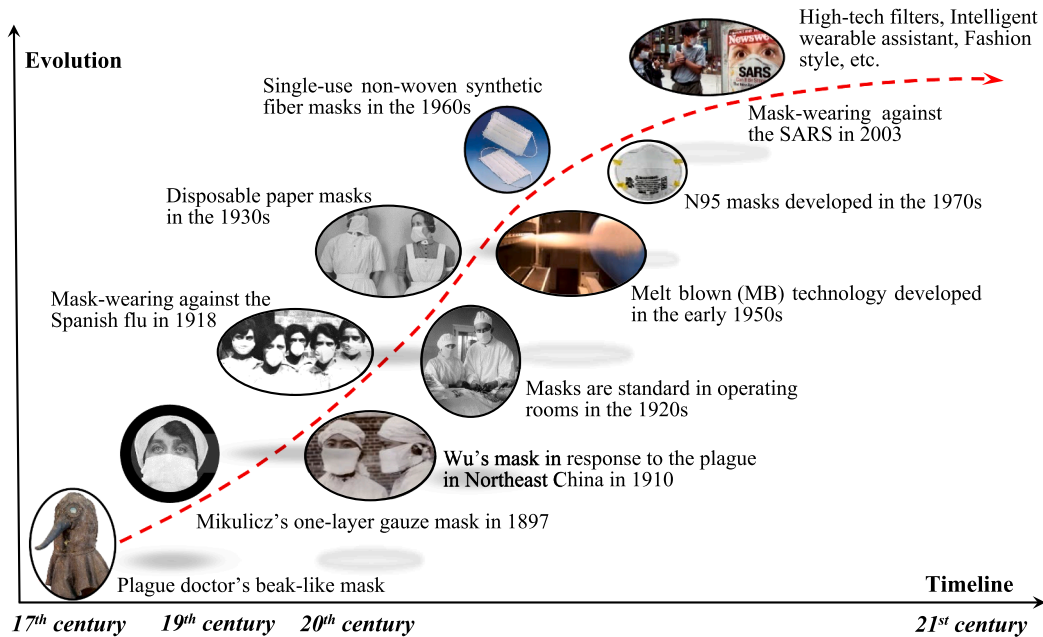


Fig. 5. S-curve of the technological evolution for medical face masks.

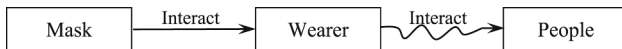


Fig. 6. Functional model of TRIZ generic problem 5 (TP 5).

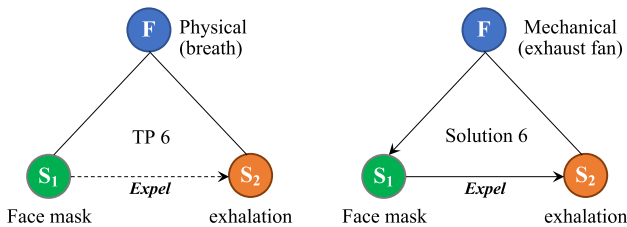


Fig. 7. Su-field analysis models for TP 6.

relative humidity, and expel trapped carbon dioxide from the mask. By using “TRIZ Inventive Principles” (Principle 1. Segmentation), the exhaust fan is an accessory to the mask designed to be attachable (Specific solution 7.1). It contains a microsensor to detect the wearer’s respiratory rate and a microactuator to adjust the fan speed (Principle 23. Feedback) (Specific solution 7.2). This accessory can be recharged via a built-in micro-USB charging port (Principle 6. Universality) (Specific solution 7.3).

According to the Inventive Principles 2, 26, 35, and 40 extracted from the contradiction matrix, TP 7 can be resolved in 3 ways as presented in Table 2. These 3 ways were combined into one solution as “The filtering composition can be attached to the mask frame by where the former is disposable or washable/sterilizable and the latter is reusable, securing the mask frame with sweat-resistant ear loops, head ties, elastic straps, or velcro bands” (Specific solution 8). In addition to the material selection, filtering composition associated with a micro exhaust fan can help improve respiratory quality so as to reduce the risk of nasal itching and skin irritation (Specific solution 7). TP 8 is concerned with the contradiction of product serviceability. According to the Inventive Principles 1, 10, 18, 32, 35, and 38 extracted from the contradiction matrix, it can be resolved in 6 ways interpreted by referring to Zhang, Chai, and Tan’s modified inventive principles (Zhang et al., 2003) and

Table 2

Inventive principles and corresponding ways of resolution for TP 7 and TP 8.

Problem	Inventive principle	Way of resolution
TP 7	2. Taking out	Separating the filtering composition from the face mask and setting up it for single-use.
	26. Copying	Designing a reusable mask frame made from skin-friendly plastics.
	35. Parameter changes	Using sweat-resistant ear loops, head ties, elastic straps, or velcro bands to secure the mask frame.
	40. Composite materials	Providing an on-line purchasing service for the filtering composition corresponding to each alternative size of the mask frame.
TP 8	1. Segmentation	Providing a pre-order online system for the customers to pre order the filtering composition.
	10. Preliminary action	Providing a just-in-time inventory system for the supply chain management of the filtering component.
	18. Mechanical vibration	Providing a revealing system that allows the customers to check the filtering composition supplies anytime and anywhere.
	32. Color changes	Integrating the above systems into a mobile app for customers to download and use.
	35. Parameter changes	Developing a timely filtering composition replacement/order reminder system to further enhance the role of customers as co-producers in the service delivery.
	38. Strong oxidants	

Gazem and Rahman’s interpretation of the 40 inventive principles (Gazem & Rahman, 2014), respectively. The 6 ways of resolution shown in Table 2 were further combined into one solution as: “The novel face mask design provides a mobile app for customers to download and use, which integrates on-line ordering/pre-ordering and just-in-time inventory revealing system with a timely filtering composition replacement/order reminder system to enhance the serviceability of the face masks” (Specific solution 9). Such O2O (online to offline) retail sales have obligations to ensure the product quality so as to reduce the risk of purchasing substandard mask filters by customers.

People around the world have paid more attention to the face mask service issues due to the outbreak of COVID-19 in late December 2019. To address domestic shortages of face masks, many countries have put in place restrictions on exports or equivalent measures such as the

compulsory purchase and distribution of all available stocks (OECD, 2020). According to the “Multi-Screen Diagram/9 Windows Analysis” shown in Fig. 8, the face mask design system includes the system itself (the problem or system), its subsystem (a component or parts of the problem or system), and supersystem (external environment and components that the problem or system interacts or may interact with). For the subsystem, the component of main materials for the active filtration layer could evolve into electrospun nanofibers due to their excellent properties of breathability and filterability (Tebyetekerwa, Xu, Yang, & Ramakrishna, 2020). Formulating the Ideal Final Result (IFR), if the face mask design aims to convert from a PPE into a fashionable wearable device, it should provide innovative system improvement in both functionality and serviceability to enhance the product value chain. Hence, the novel face mask design combines a personal breath health monitoring system (PBHMS) (Future Supersystem) with a fashionable wearable device (Future System) (Specific solution 10). The hardware of the PBHMS is integrated into the micro exhaust fan. It uses an ultrasonic sensor and microcontroller system to simultaneously monitor the wearer’s normal and abnormal breathing activity in real-time (Al-Naji, Al-Askery, Gharghan, & Chahl, 2019; Arlotto, Grimaldi, Naeck, & Ginoux, 2014) (Specific solution 10.1). Such abnormal breathing syndromes, including tachypnea, bradypnea, central apnea, and irregular breathing, are detected by the ultrasonic radar via tracking the thorax movement resulting from respiratory rhythms. All the detected data are transmitted simultaneously as big data for cloud computing by the mobile app installed on the wearer’s smartphone (Specific solution 10.2), and the wearer can immediately receive the biofeedback information of his/her exhaled breath analysis (Lawal, Ahmed, Nijssen, Goodacre, & Fowler, 2017) (Specific solution 10.3). This would be beneficial for early detection and early treatment of abnormal breathing syndromes.

On the basis of the derived specific solutions, the face mask can be divided into 6 parts: mask frame, filtering composition, secure manner, micro exhaust fan, LED indicator, and microphone. It also includes auxiliary service, value-added service, and mobile app download for service offering. According to the morphological analysis shown in Table 3, there are theoretically 1,728 ($3 \times 3 \times 3 \times 4 \times 4 \times 4 = 1,728$) sets of concept possibilities in the solution space of product design,

created by permutations and combinations of these option parameters. Through a preliminary assessment, the possible configurations were reduced to 8 alternative design concepts by eliminating some inappropriate parameters. For example, the parameters of microphone (S6) were excluded as they can induce noise by interference from the running of the exhaust fan. The combinations of option parameters and idea sketches for the 8 alternative design concepts are shown in Fig. 9. The service design can also be determined by permuting and composing the option parameters. These can be either free services or fee-based services depending on the service providers’ competencies and capabilities or their established business models.

4.4. Concept evaluation

To operate the linguistic variables consistently, the rating scales for negative items (i.e., sub-criteria of costs c_{11} , c_{12} , and c_{13}) were reversed, so that lower costs indicated higher performance. Table 4 presents the fuzzy linguistic evaluation results. According to the prioritizing results, the most desirable concept for the novel face mask design is D3. Hence, it is selected as the promising concept for further engineering and manufacturing development.

4.5. Embodiment design

According to the idea sketch of D3, the embodiment design was undertaken using 3D computer representations, and then the mockup of the novel face mask was fabricated, as shown in Fig. 10, which is used for examining the following engineering and manufacturing procedures. The mobile user interface design was also created to provide users with access to acquiring service applications, considering users’ limited attention, minimizing keystrokes, maximizing usability and the UX, and being task-oriented with a minimum set of functions (see Fig. 11). The PBHMS was renamed to “Personal breath health management system” (also abbreviated as PBHMS) as it is used not only to monitor the breathing parameters but also to provide breathing health management services. In addition to the face mask sales, the fee-for-service business model includes daily O2O on-demand delivery of filter consumables for the face mask usage and annual membership for using the PBHMS.

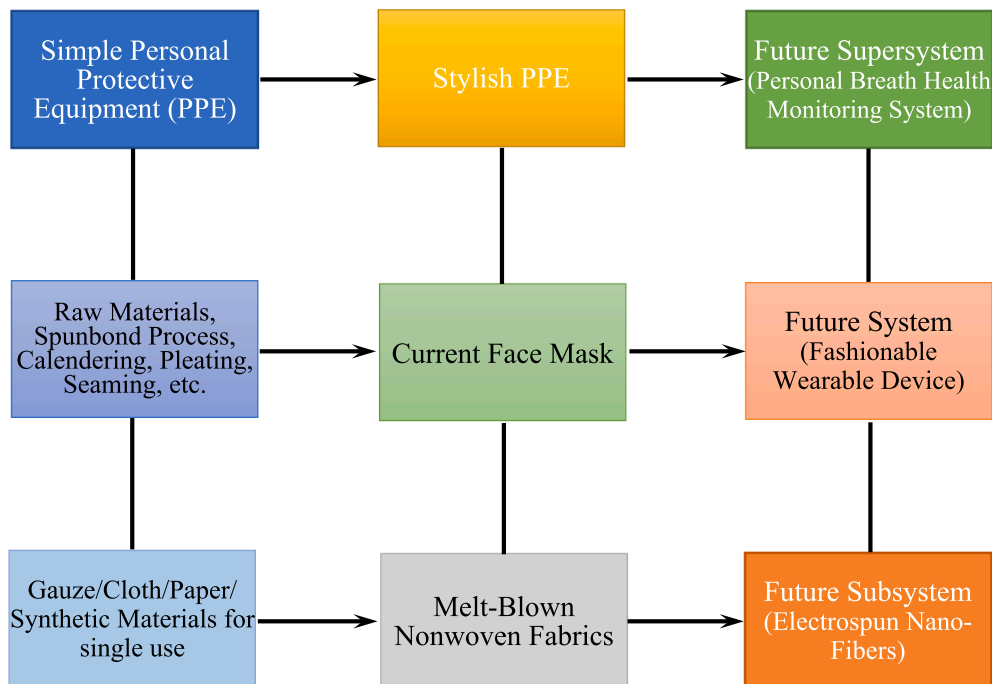


Fig. 8. Multi-screen diagram/9 Windows analysis for the face mask design.

Table 3

Morphological chart for concept generation.

Product design	Option parameter (concept solution)			
	Option 1	Option 2	Option 3	Option 4
S1: Mask frame	Half face mask	Simple face mask	Other type in shape	
S2: Filtering composition	Disposable	Reusable/Washable	Replaceable	
S3: Secure manner	Earloop	Headband	Neck-hanging	
S4: Micro exhaust fan	Attached to the mask frame	Attached to the filtering composition	Fixed on the mask frame	Covered by the filtering composition
S5: LED indicator	Embedded in the mask frame	Embedded in the micro exhaust fan	Attached to the filtering composition	None
S6: Microphone	Embedded in the mask frame	Embedded in the micro exhaust fan	Attached to the filtering composition	None
Service design	Option parameter			
	Option 1	Option 2	Option 3	
Auxiliary service	AQI information	Instant translation	Weather information	
Value-added service	Ordering system	Personal breath health monitoring system	Emergency assistance	
Mobile app (for download)	App Store	Google Play	Others	

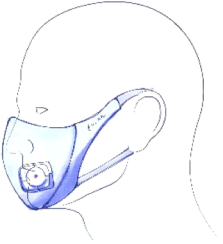
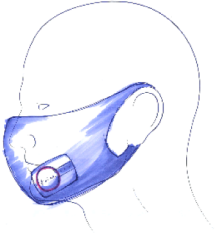
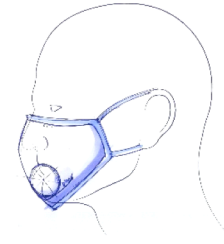
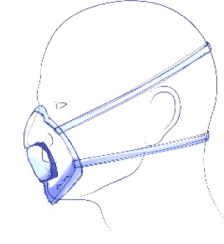
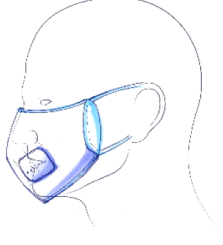
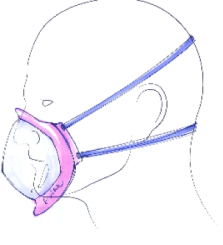
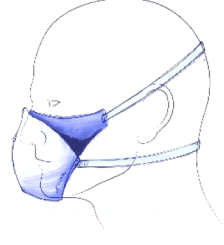
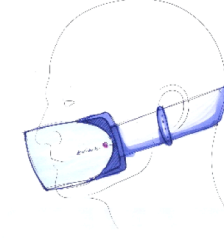
D1: Design concept 1	D2: Design concept 2	D3: Design concept 3	D4: Design concept 4
			
S1.1+ S2.3+ S3.1+ S4.2+ S5.2+ S6.4	S1.1+ S2.2+ S3.1+ S4.2+ S5.2+ S6.4	S1.1+ S2.1+ S3.1+ S4.1+ S5.2+ S6.4	S1.2+ S2.3+ S3.2+ S4.3+ S5.1+ S6.4
D5: Design concept 5	D6: Design concept 6	D7: Design concept 7	D8: Design concept 8
			
S1.1+ S2.1+ S3.1+ S4.2+ S5.2+ S6.4	S1.1+ S2.1+ S3.2+ S4.4+ S5.2+ S6.4	S1.2+ S2.3+ S3.3+ S4.3+ S5.1+ S6.4	S1.3+ S2.3+ S3.2+ S4.3+ S5.1+ S6.4

Fig. 9. Combinations of option parameters and idea sketches for the eight alternative design concepts.

4.6. Product and service specifications

According to the final design and development results, the product and service specifications are developed as follows:

4.6.1. Product specifications

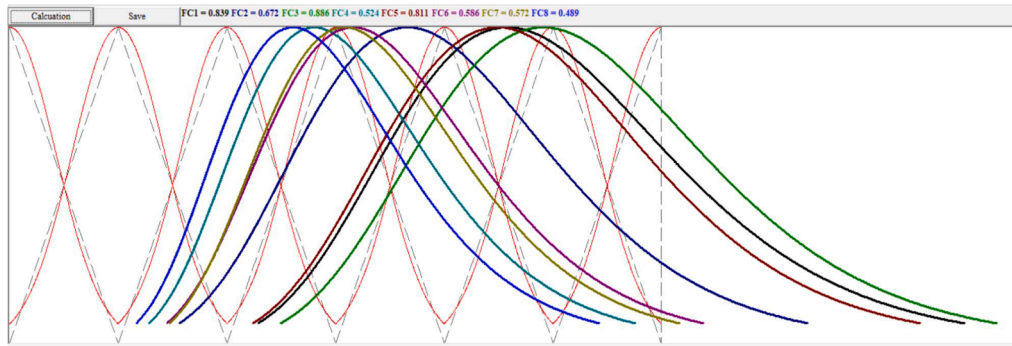
1. The product design combines a half face mask with a wearable device, which mainly consists of a flexible mask frame and disposable mask filters, providing wearers with fashionable and stylish mask-wearing experience.

- The mask frame is made from a super-elastic memorized resin (abbreviated as TR 90, also known as plastic titanium) and secured to the head with sweat-resistant ear loops. It is used to uphold the disposable mask filters and provides 4 different sizes (S, M, L, and XL) for fitting different wearers.
- The disposable mask filters are made of non-woven fabrics with different kinds of active filtration layer for different usage situations, such as melt-blown polymer and electrospun nanofibers for stopping microbes, N95 grade polyester for filtering small airborne particles, activated carbon filters for trapping allergens, and photocatalytic filters for destroying microparticles.

Table 4

Fuzzy linguistic evaluation results for the alternative design concepts.

Criterion	Sub-criterion	Weigh	Rating of alternative concept design							
			D1	D2	D3	D4	D5	D6	D7	D8
C ₁ : Economic viability	Manufacturing costs (c ₁₁)	M	M	VH	H	L	M	L	L	VL
	Assembly/packaging costs (c ₁₂)	M	M	VH	H	VL	MH	L	VL	L
	Maintenance costs (c ₁₃)	MH	H	H	MH	L	M	M	ML	L
C ₂ : Social acceptability	Quality of life (c ₂₁)	VH	H	ML	H	MH	H	MH	H	MH
	Health and well-being (c ₂₂)	H	H	M	H	MH	H	MH	MH	MH
	Safety and security (c ₂₃)	VH	H	L	VH	H	VH	H	H	H
C ₃ : Environmental soundness	Materials used (c ₃₁)	ML	H	VH	MH	L	H	ML	L	L
	Energy saving (c ₃₂)	M	MH	H	MH	ML	MH	ML	ML	L
	Waste reduction (c ₃₃)	MH	H	MH	H	M	MH	M	M	ML
Quantitative value/Rank			0.839/2	0.672/4	0.886/1	0.524/7	0.811/3	0.586/5	0.572/6	0.489/8



Algorithmic results via the developed G-FWA computer program

**Fig. 10.** Physical embodiment of the novel face mask design.

4. A micro exhaust fan is attached to the mask frame to expel exhalation according to the wearer's respiratory cycle. It contains a microsensor to detect the wearer's respiratory rate and a microactuator to adjust

the fan speed. The micro exhaust fan can be recharged via a built-in micro-USB charging port.

5. An annular LED indicator is embedded in the micro exhaust fan to display color signals according to the corresponding air quality index

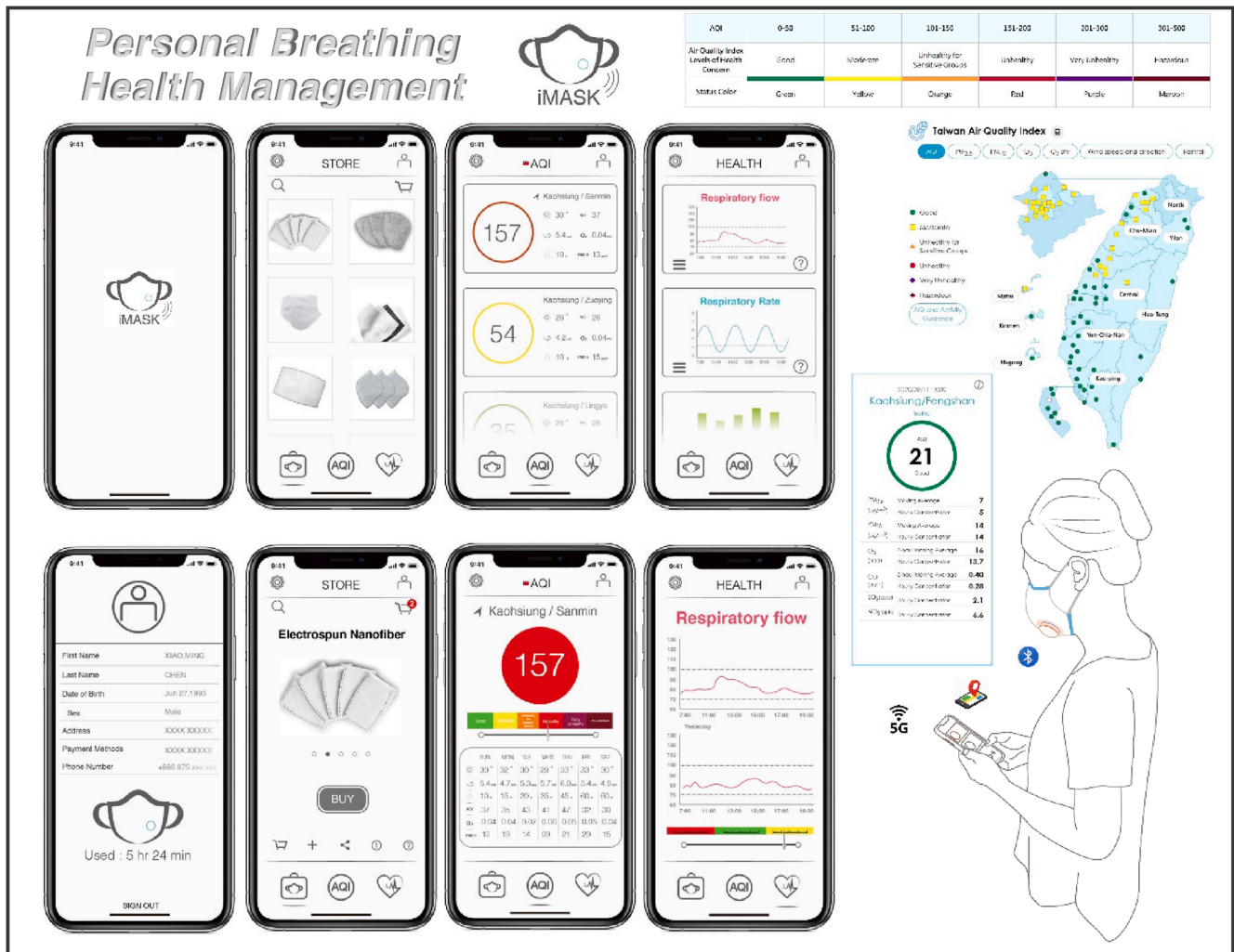


Fig. 11. User interface design for mobile applications.

(AQI) scale levels by the wearer's immediate location to remind people around him/her. The AQI data is derived from the wearer's smartphone through the GPS and Bluetooth connection, and Mobile Communication Networks (MCN).

6. An ultrasonic sensor system with microcontrollers is also embedded in the micro exhaust fan to simultaneously monitor the wearer's breathing activity. All the detected data are transmitted immediately as big data for cloud computing by the mobile app installed on the wearer's smartphone.

4.6.2. Service specifications

1. A mobile app “i-Mask” is provided for customers to download and install on their smartphones (for both Apple’s iOS and Google’s Android), which mainly consists of Mask Filter Ordering (MFO) system (STORE), Air Quality Index system (AQI), and Personal Breath Health Management (PBHM) system (HEALTH).
2. The MFO system (STORE) contains on-line order/pre-order, just-in-time inventory revealing, timely filter replacement notice, and purchase reminder. The system allows customers to choose their preferred delivery and payment methods.
3. The AQI system (AQI) enables customers to know how clean or polluted the immediate location-specific air is. It describes air quality levels by color codes and each color corresponds to a different level of health concern. The AQI information is officially reported by

government agencies in a national standard, and the system usage is free of charge.

4. The AQI system (AQI) also provides GPS-located weather information including temperature/apparent temperature, relative humidity, probability of precipitation, and ultraviolet index.
5. The PBHM system (HEALTH) provides ordinary members with basic breath monitoring data for free, such as respiratory rate, respiratory flow, and respiratory rhythm. If customers upgrade as members and paid the annual membership fees, they can regularly receive their own private information of exhaled breath analysis. Such biofeedback information is used for building respiratory disease database according to demographic data like gender, age, height, weight, etc. Each member allows to compare the detected respiratory data with his/her own historical records and instantly receive the AI-based diagnostic analysis and assessment results. The use of the PBHM system can help users early detect and early treat respiratory diseases such as asthma, bronchitis, pneumonia, lung cancer, and chronic obstructive pulmonary disease (COPD).
6. Since air pollution contributes to respiratory diseases, the PBHM system can also recommend the proper type of mask filters for users to order and use according to the user's breathing status and everyday surroundings.

4.7. Validating the results

Face masks are an efficient PPE not only for against the COVID-19

pandemic but also for implementing personal hygiene and public health. Traditional face masks or surgical masks have a profound image of medical device and can make users perceive certain degree of discrimination and stereotyping regarding mask-wearing. This product design case integrated a face mask with a wearable device, providing users novel functionality and serviceability with unique usability and wearing experience. The developed face mask is a cyber-physical product that can detect, analyze, and transmit a wearer's breathing data in real time and give the corresponding biofeedback and internal/external environment information to the wearer. It also provides a mobile app to optimize the operations and usage of the face mask, enabling useful functions and services of the product to perform beyond the physical device.

The user interface design mainly focused on developing the mobile app we called "i-Mask", which consists of MFO (STORE), AQI, and PBHM (HEALTH) systems. Each system allows users to intuitively and consistently interact with the on-screen content or objects via their smartphones. The product lifecycle design aimed to improve the sustainability of face masks by taking environmental considerations in each life cycle stage of the PDD processes. The mask frame with ear loops is reusable and are fabricated using recyclable materials. For hygiene and security reasons, the mask filters are disposable and can be replaced with different kinds of active filtration layer for different usage situations. The attachable micro exhaust fan is rechargeable, making its usage more energy-efficient and environment-friendly. In addition to the design patent application, the design proposal was submitted to the Emerging Bright Industry Co., Ltd and the LSH Electronics Co., Ltd for further production assessment. The results indicated that the proposed TRIZ-based product-service design approach is applicable to designing innovative cyber-physical products.

4.8. Limitations of the case study

The proposed approach is applicable to generating novel design concepts and identifying promising concepts; however, a number of questions regarding the effectiveness and limitations of the case study remain to be addressed. The proposed approach requires designers/engineers to convert the problems from factual domains to TRIZ domains, and then integrate the TRIZ solutions into concrete product and service concepts. However, individuals may have different quantity and quality of domain-specific knowledge, and also have different capacities of analogical reasoning for seeking ways to convert the TRIZ guidelines/principles/parameters into concrete product and service concepts. Moreover, individuals' expertise and experience may influence the determination of promising concepts in such decision-making context as intuitive linguistic judgements.

Another limitation is concerned with the embodiment design. It is a complex process as many design activities must be performed concurrently. Most design processes inevitably require iterative modifications from time to time to fit the PDD goal. They are closely linked and mutually affected, and require effective collaboration, coordination, and cooperation among work teams. In this design case, the attachable exhaust fan is a key component for the face mask. It is an innovative accessory that integrates micro exhaust fan, annular LED indicator, ultrasonic sensor, microcontroller, Bluetooth module, and built-in rechargeable battery into a plastic casing. How to develop an integration of the microchip and micro electro-mechanical system (MEMS) is quite an engineering challenge, which requires investing significant amounts of funds.

5. Discussion

On the basis of the 4 pillars of functionality, serviceability, usability, and sustainability, this research presents a TRIZ-based product-service design approach to assist designers/engineers in developing innovative products. An empirical study was conducted to verify the applicability

and validity of the proposed approach, the purpose of which was to develop a novel face mask product for use against the COVID-19 pandemic. The developed product can address the high innovation level of socio-technical system to promote radical changes in what the society needs to change, as mentioned earlier by [Chechurin and Borgia \(2016\)](#). The proposed approach has certain advantages over other conventional design methods and PSS models. The implications for theory and practice of the proposed approach are discussed below.

Conventional design methods deal with products and services by distinct perspectives and using specific techniques, making it difficult to integrate both into a holistic design process. Early CE paved the way for interdisciplinary integration and collaboration in PDD processes, while current PSSs envision a future in coupling products and services to co-create more satisfying value for customers. There have been various frameworks for supporting PSS design in the literature (e.g., [Cavaliere & Pezzotta, 2012](#); [Fargnoli & Haber, 2019](#); [Muto, Kimita, & Shimomura, 2015](#); [Trevisan & Brissaud, 2016](#); [Vasantha et al., 2012](#)), but existing PSS frameworks are mainly concerned with general PSS and service design (e.g., product-oriented services, use-oriented services, and result-oriented services), which cannot readily assist designers/engineers in developing consumer products to implement and realize PSS solutions ([Yang, Moore, Pu, & Wong, 2009](#)). In recent years, Smart PSS has evolved from conventional PSS to integrate state-of-the-art product technologies and e-services into a value co-creation system, providing integrative insights to develop ICT-based product-service system with less environmental impact. Although the theoretical constructs of PSS and Smart PSS are not central to dealing with service development only, recent studies, particularly empirical research with practical cases, still focus on cyberspace operations for service innovations, such as smart beauty service ([Lee, Chen, & Trappey, 2019](#)) and the digitalization of restaurant service ([Watanabe et al., 2020](#)). [Zheng et al. \(2018\)](#) proposed a systematic design framework and elaborated the framework via a case study of a personalized smart wearable they called "i-BRE respiration mask"; however, this case study mainly aimed at data-driven design for service innovation even though their proposed framework involves both cyber space and physical space. The reasons why current research concentrates on cyber service innovations and neglects the physical product development might be due to the diverse definitions of Smart PSS, which are less consistent and primarily depend on what contents and which lifecycle stages the researchers take. To bridge the gap, this research offers new perspectives on cyber-physical products (CPP) defined as a physical product combined with its ICT-enabled service support to fulfill customers' requirements while minimizing environmental impact.

Of the proposed approach, concept generation and concept evaluation are two critical processes for deriving various design concepts and further identifying promising ones. TRIZ is capable of dealing with problems of product, service, usability, and sustainability by means of pattern-based guidelines to solve different types of conflict problems and generate novel design concepts. Although applying TRIZ to smart PSS design has been presented in the recent research literature ([Lee et al., 2019](#)), it still aims at resolving service problems rather than the entire product and service problems. TRIZ provides a variety of analytical and knowledge-based tools to assist designers/engineers in finding and formulating problems as well as generating solutions. These can help them converge numerous generic solutions into a rational range of specific solution space to develop their product and service concepts. On the basis of TRIZ advantages over traditional innovation support methods, this research developed a TRIZ-based design approach and conducted a case study to elaborate how to apply the developed approach to designing an innovative cyber-physical product. Although TRIZ has been recognized as a powerful methodology and successfully employed in industry and academia for over half a century, the complex nature of TRIZ methodology, including its philosophy, usage, and tool-kit, makes it difficult for unexperienced users to use. Besides, there has been criticism in the TRIZ community that the generic solutions suggested by TRIZ guidelines/principles/parameters are abstract rules of

thumb and can confuse designers/engineers in the operations from abstraction to concretization (Mann, 2002). The interconversion between factual domains and TRIZ domains can facilitate the problem-solving performance; however, the conversion from TRIZ solution space to concrete solution space is quite a challenge that requires high levels of analogical association. How to integrate and convert the corresponding TRIZ generic solutions into concrete product and service concepts is of vital importance to ensure success in the concept generation process.

Conventional design methods primarily highlight the importance of problem solving and concept generation, and place less emphasis on the effective decision-making methods for the concept evaluation procedure. PSS development and implementation require effective evaluation (Mourtzis, Papatheodorou, & Fotia, 2018; Nakada, Sholihah, Mitake, & Shimomura, 2020), and there have been a number of techniques for evaluating PSS in the literature, such as the analytic network process (ANP) and data envelopment analysis (DEA) for prioritizing engineering characteristics of PSS (Geng, Chu, Xue, & Zhang, 2010), the rough DEMATEL-based approach for evaluating interaction between PSS requirements (Song & Cao, 2017), etc. To effectively capture human perception involving inherently subjective, uncertain, and vague characteristics, fuzzy evaluation models have been applied to PSS/Smart PSS over the past decade. For example, Sousa-Zomer and Miguel (2017) applied fuzzy AHP to prioritizing the stakeholders' requirements for the conceptual design of a bike-sharing system. Liu et al. (2020) further evaluated the conceptual factors of smart PSS by using a hybrid approach of integrating fuzzy DEMATEL, modified-ANP, and fuzzy TOPSIS. Although their approach can improve the drawback of ignoring inherent relevance among PSS factors in the context of using fuzzy AHP, it is complicated and rather time-consuming for evaluators to perform, such as the pairwise comparisons with linguistic scales for fuzzy DEMATEL, the construction of fuzzy direct-relation matrix, the fuzzy TOPSIS operations for prioritization processes, and the sensitivity analysis for checking the reliability of the prioritization results. Chen, Ming, Wang, and Bao (2020) also proposed a rough-fuzzy DEA method to select the proper smart PSS design alternatives using a 5-point scale defined by triangular fuzzy numbers. Although the α -cut-based triangular fuzzy DEA method can be readily used to rank alternatives, it does not use weights to represent the relative importance of criteria. The generic term criteria weights/importance is a key element for any multicriteria decision making (MCDM) model (Choo, Schoner, & Wedley, 1999; Odu, 2019), particularly with respect to such PSS design evaluation involving a hierarchical structure of economic, social, and environmental criteria. This research presents a fuzzy weighted average (FWA) method with linguistic variables to evaluate alternative design concepts. It uses a 7-point scale defined by Gaussian fuzzy numbers as rating (preference) and weight (importance) variables to intuitively prioritize alternatives. As Gaussian membership functions (GMFs) can be statistically considered a normally-distributed ordered set of rating scales, they are more suitable for problems requiring continuously differentiable curves to capture the vagueness of rating responses as compared with the most commonly used triangular membership functions (TMFs). Hence, the proposed approach is capable of capturing evaluators' appraisals of ambiguity and allows them to incorporate unfit, deficient, and unavailable information, and partially unknown facts into a decision-making process.

Appendix A. ARIZ operation for TRIZ generic problem 1 (TP 1)

6. Conclusions

This study attempted to address the shifting interest of research and practices in Smart PSS from the viewpoint of product design and development (PDD). It briefly analyzed the contexts of PDD evolution and categorized the developed strategies, theories and methodologies in the existing literature. A TRIZ-based product-service design approach was proposed to assist designers/engineers in developing innovative products. In the context of COVID-19 pandemic, an empirical study concerning a novel face mask design was also conducted, which showed that the proposed approach is applicable and effective to facilitate the design and development of innovative cyber-physical products.

In conclusion, this study contributes to our knowledge by improving PDD processes and enhancing the strength of PSS/Smart PSS design implementation. Academically, this study offered new perspectives on cyber-physical product (CPP) design beyond the developing Smart PSS design. The framework of the proposed approach provides a new design paradigm based on the 4 pillars of functionality, serviceability, usability, and sustainability. In practice, this study proposed a novel design approach for designers/engineers to develop innovative CPPs, aiming to achieve the total performance of products and services while minimizing environmental impact. Although TRIZ provides powerful tools for solving product and service problems, it requires a high-level capacity to perform analogical reasoning for converting the TRIZ solutions into concrete design concepts. Structure mapping is a theory that explains an analogical mapping of knowledge from one base domain into another target domain. It posits that schema analogues can be viewed as being similar according to their relational structures or how they relate. Future research could focus on developing a TRIZ-based structure mapping approach to assist designers/engineers in transferring their relevant domain knowledge into specific target domains, and determining the best correspondences between schema and relations in TRIZ solution domains and those in design concept domains.

Credit Author Statement

The author confirms sole responsibility for the following: study conception and design, data collection, analysis and interpretation, drafting the article, critical revision of the article, and final approval of the version to be published.

Three students (Ko-Hsun Hsu, Yueh-Mei Li, and Chun-YU Chou) contribute to the embodiment design of the face masks in the illustrative case study, including idea sketches, 3D modelling, mockup making, and UI design. They are my paid assistants under the research grant numbered MOST 108-2221-E-214-003.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Problem description:

A system for *<providing wearers with a physical barrier in public>* consists of *<a filtering composition, ear loops, and nose bridge>*.

Contradiction formulation:

The mini-problem is formulated as:

Technical contradiction 1 (TC-1):

If there is *<a filtering composition>*, then *<it covers your face to protect you from potential contaminants in the immediate environment>*, but *<it changes the way you look which may cause negative feelings around your identity or body image>*.

Technical contradiction 2 (TC-2):

If there is *<no filtering composition>*, then *<there is no related change of your identity or body image>*, but *<you are not protected from potential contaminants in the immediate environment>*.

Desired result:

It is necessary, with minimal changes to the system, *<to protect you from potential contaminants in the immediate environment with a filtering composition without changing the way you look>*.

Problem model:

(1) Conflicting pair:

<filtering composition> and *<the way you look>*

(2) Intensified conflict:

<In addition to the filtering composition, the face mask> requires *<ear loops to tighten the mask and nose bridge to fit your face>*, but *<it also causes you feel uncomfortable while wearing a mask>*.

(3) Problem:

It is necessary to find an X-element which maintains the ability of *<providing wearers with safe and convenient protection>* and minimizes *<negative feelings>*.

Definition of the operational zone (OZ) and operational time (OT)

The operational zone (OZ) and operational time (OT) are defined as “*the wearer is wearing a face mask (OT) in public (OZ)*”. The improvement opportunity could focus on changing the face mask’s product image and usage to eliminate the conflict indicated in the problem model.

Formulation of the ideal final results (IFRs) and physical contradictions (PhCs)

IFR-1:

The X-element, without complication of the system and without harmful side effects, eliminates *<the negative feelings of the wearer>*, and keeps *<the face mask's>* ability to protect *<the wearer with safe and convenient protection>* during the *<operational time (wearing)>*, inside the *<operational zone (in public)>*.

Intensified IFR-1:

<A new element of changing the face mask design with a whole new product image> eliminates *<the negative feelings of the wearer>* inside the *<operational zone (in public)>*, during the *<operational time (wearing)>* and protects *<the wearer from potential contaminants in the immediate environment>* without complicating the system and without harmful side effects.

PhC for macro-level:

<The new element in the operational zone (in public)>, during the *<operational time (wearing)>* has to *<provide face masks with a whole new product image>* in order to avoid *<the negative feelings for the wearer>* and has to *<be used as a filtering composition >* to protect *<the wearer from potential contaminants in the immediate environment>*.

PhC for micro-level:

There should be contained *<an adjustable nose bridge>* in the operational zone (OZ) in order to provide *<wearers with ease of fitting>* and there should be contained *<elastic ear loops>* during the operational time (OT) in order to protect *<wearers from feeling uncomfortable>*.

IFR-2

<The new element in the operational zone (in public)> has to *<better the product usage>* during the *<operational time (wearing)>*.

Proposed solution:

The product image and product usage are available resources within the face mask design system. Product image refers to the perceptions and attitudes of customers toward an existing or proposed new product, while product usage is concerned with how or when users interact with the product. Wearable devices are new manifestations of accessories that combine aesthetics and style with functional technology for people to wear. Such devices can be used as the existing resource for the system change and resolution development of mask-wearing. According to “TRIZ Separation Principles” (Separation upon condition), the face mask design should be separated from its conventional image (a medical device) and usage (wearing a personal protective equipment, PPE). By using “TRIZ Inventive Principles” (Principle 3. Local quality), the face mask design is changed from a traditional medical device to a modern wearable device.

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